

UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Homework 4: Read/Write Files and Probability
Due Monday, September 21, 2:30pm

Goals of Practicum 2 and this homework:

Become sufficiently familiar with Python to:

1. Read, parse, and write ASCII data files

Become sufficiently familiar with probability and statistics to:

1. Calculate the mean, median, mode, variance, and standard deviation of a group of measurements.
2. Understand the Gaussian and Poisson distributions, their parameters, and their application in astronomical observation.
3. Memorize the formula for the Gaussian distribution.

Resources: Chapters 1 – 4 and 6 of Bevington

1. **(5 points)** Use your terminal to make a new folder `hw4_yourname` under your `ast4762/homework` folder. Explain in your log how you did it. For this homework your main homework file is a Python file named `hw4_<yourname>.py` (or `hw4_<yourname>.ipynb` but remember to save it as a `.py` before submission; see end of homework, too). Save it in your homework (`hw4/`) folder. Remember to commit your files and push to GitHub as often as needed (also, great backup!).

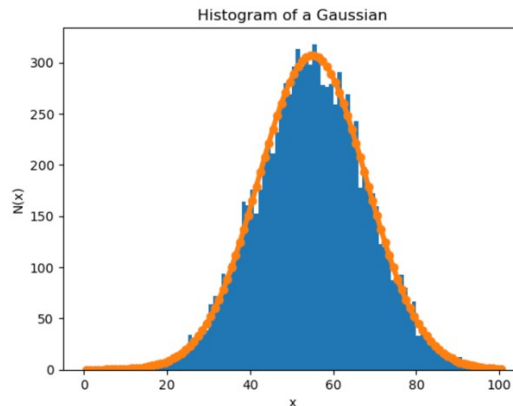
Your name, assignment number, and the date should appear as comments at the top of the file. At the start of every problem, print the problem number using the appropriate Python command. Be sure to follow good coding habits throughout your code!

2. The Gaussian distribution in Python **(30 points total)**

- a. **(10 points)** Find a Python function that draws a random sample of N draws from a Gaussian distribution of width σ and mean μ . Use it to make a sample of 10,000 draws (i.e., $N=10,000$) from a population with $\sigma = 13$ and $\mu = 55$.
- b. **(10 points)** Plot the histogram of your sample. The histogram should have bins from $x = 0$ to 100 with a width of 1 unit. Be sure to make a publication-ready plot (with axis labels and a title). Save it as a PNG file using the appropriate Python command. Remember to name your PNG appropriately.
- c. **(10 points)** Now use the Gaussian function to overplot a Gaussian distribution with the same μ and σ as your histogram. Approximate the integration between the bin boundaries by just evaluating the Gaussian at the center of each bin (e.g., for bin $[1,2]$ you will evaluate the Gaussian at $x = 1.5$). Remember that the analytic Gaussian distribution, integrated over each bin, gives you the expected number of draws in each bin for a *single* draw from the parent population (i.e., much less than one draw per bin on average, in contrast

to your histogram from parts a and b). If you thus try to plot your function “as is” versus the histogram you will not see them aligning properly. To fix this, you need to multiply your function by the total number of draws N to get the expected number of draws per bin for a larger sample and get the correct plot.

Save the new plot as a PNG file. At the end you should get a plot like this one:



3. **AST 5765 only, extra credit for AST 4762 (20 points total):**

- a. **(10 points)** Show analytically that the full width at half maximum (FWHM) of a Gaussian distribution is 2.354σ (See Bevington, p. 28). This can be done in a PDF file you make, a photo of a solution written on paper, or in comments in your homework file. If it's in a PDF file or image, please refer to the filename in your homework file.
- b. **(10 points)** A line on a log-log plot has a specific meaning, and is the main purpose of log-log plots. Derive the relationship of x , y , and the slope and intercept of a line in a log-log plot.

4. **(5 points)** Test that your code works in Stokes and submit your homework. If you worked in a Jupyter notebook, make sure to save it as a .py file. Then:

- a) Log in to Stokes. Explain how you did this in your log.
- b) Make a folder `hw4_<yourname>`. Explain how you did this in your log.
- c) Copy your `.py` file into the folder. Explain how you did this in your log.
- d) Test it! Does it run without issues?
 - a. Yes! Great. Move on to the next steps
 - b. No. Debug!
- e) Don't forget to commit and push to GitHub. Explain what you did to do that in your log.
- f) Make a screenshot that shows your logging history, showing that you committed the file and add it to your `hw4_<yourname>` folder (remember to use an appropriate name for the screenshot! Also, don't forget to copy it to your Stokes folder, too).

- g) If needed, copy the rest of the content of your `homework/hw4_<yourname>` folder into your Stokes folder (e.g., any screenshots, etc., that might have been requested).
- h) Explain how you will do the following in your log and then copy the log into your Stokes folder:
 - a. Copy the log into the Stokes folder.
 - b. Make a `.tar.gz` of the folder on Stokes and explain how you did this in your log.
 - c. Download the `.tar.gz` file to your local `handin/` folder.
 - d. Submit the file to Webcourses.
- i) Copy your log into your Stokes folder and `tar.gz` your file.
- j) Download your `.tar.gz` and submit your homework.